

KONGSVINGER, Norway

WMO: 014680

Lat: 60.1903N

Lon: 12.0067E

Elev: 486

StdP: 14.44

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.4	(c) -0.1	(d) -9.9	(e) 3.3	(f) -6.3	(g) -3.2	(h) 4.7	(i) 0.2	(j) 16.1	(k) 39.6	(l) 13.7	(m) 36.9	(n) 2.9	(o) 100	(p) 0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	80.6	62.4	76.8	60.8	73.0	59.6	64.8	75.1	63.1	72.5	61.4	69.6	4.5	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.2	82.1	67.6	59.5	77.1	66.1	58.0	73.0	64.6	30.1	74.8	28.8	72.6	27.6	69.7	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	10.8	9.3	-10.8	85.0	6.2	4.1	-15.2	88.0	-18.9	90.4	-22.3	92.7	-26.8	95.6
			-12.0	67.3	6.7	2.4	-16.8	69.0	-20.7	70.4	-24.5	71.8	-29.3	73.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.2	21.4	25.3	32.5	41.4	50.6	58.0	62.5	59.2	52.3	41.6	33.5	27.0
	DBStd	15.59	11.62	9.45	7.80	6.17	6.99	5.30	5.21	4.11	5.14	6.23	7.65	10.28
	HDD50	3978	885	692	541	267	78	3	0	0	34	270	494	714
	HDD65	8372	1350	1112	1006	707	448	219	115	186	380	726	944	1179
	CDD50	1134	0	0	0	9	96	243	388	285	104	9	0	0
	CDD65	53	0	0	0	0	1	9	38	5	0	0	0	0
	CDH74	642	0	0	0	0	58	158	394	32	0	0	0	0
	CDH80	116	0	0	0	0	3	22	91	0	0	0	0	0
Wind	WSAvg	4.1	4.0	4.0	4.2	4.4	4.4	4.3	4.0	3.9	3.8	3.8	4.0	4.3
Precipitation	PrecAvg	28.5	1.9	1.4	1.3	1.9	2.1	3.1	3.1	3.5	2.5	3.0	2.7	2.0
	PrecMax	40.5	3.3	2.8	3.7	3.8	5.4	5.2	4.7	5.9	4.4	7.7	7.4	3.9
	PrecMin	24.2	0.4	0.4	0.0	0.1	0.3	0.7	0.9	0.6	0.9	0.9	0.9	0.4
	PrecStd	4.1	0.9	0.7	1.0	1.0	1.4	1.3	1.1	1.5	1.1	1.6	1.7	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.1	44.8	58.4	67.4	79.1	83.3	86.5	77.7	71.3	60.4	51.9	49.0
		MCWB	41.0	39.4	46.2	51.0	58.8	61.1	65.0	62.8	60.0	53.9	48.7	46.1
	2%	DB	40.8	41.4	51.9	63.7	75.1	78.6	82.4	73.9	67.0	56.4	49.1	45.7
		MCWB	38.5	37.8	42.3	49.9	56.8	59.6	63.8	61.1	57.4	51.6	46.5	43.1
	5%	DB	38.4	39.2	46.5	59.1	70.0	74.6	78.5	70.8	63.7	53.8	46.7	42.9
		MCWB	36.5	36.7	39.0	47.3	55.2	58.6	63.4	59.6	56.0	49.6	44.4	40.8
	10%	DB	35.3	36.7	43.0	54.6	64.8	69.9	74.3	67.9	61.1	51.2	44.1	40.1
		MCWB	33.6	34.7	37.7	44.9	52.8	56.2	61.3	58.5	54.6	47.9	42.3	38.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.8	41.0	47.1	53.1	61.9	63.9	68.3	65.4	62.0	55.7	49.5	46.5
		MCDB	43.5	43.1	57.2	63.2	72.5	73.7	80.0	71.4	66.0	58.6	51.3	48.7
	2%	WB	39.0	38.6	43.0	50.6	59.0	61.9	66.0	63.4	59.3	53.1	46.8	43.2
		MCDB	40.4	40.6	50.5	61.0	70.7	73.6	77.4	69.9	63.8	55.4	48.7	45.1
	5%	WB	36.5	36.8	40.3	48.4	56.5	59.9	64.3	61.7	57.4	50.5	44.6	40.8
		MCDB	38.1	38.7	45.0	57.7	66.9	71.7	75.9	68.1	62.6	53.1	46.3	42.5
	10%	WB	33.9	34.9	38.3	45.8	54.1	58.0	62.5	60.0	55.5	48.3	42.5	38.0
		MCDB	35.1	36.5	42.4	53.6	63.2	68.1	72.2	65.7	59.9	50.7	44.1	39.7
Mean Daily Temperature Range	5% DB	MDBR	10.5	12.1	16.6	19.4	18.7	17.9	17.1	15.4	15.3	12.2	8.8	9.4
		MCDBR	10.1	13.5	22.5	27.6	28.8	26.5	24.7	20.2	19.4	15.2	12.0	11.4
		MCWBR	9.1	10.2	13.8	14.9	12.9	10.7	10.0	9.8	11.2	10.4	10.2	10.2
	5% WB	MCDBR	9.8	11.5	20.0	25.1	23.8	22.2	22.1	16.9	16.0	12.3	11.3	11.1
		MCWBR	9.0	9.2	12.8	14.0	11.7	10.3	9.5	9.5	9.9	9.5	10.1	10.2
Clear-Sky Solar Irradiance	taub		0.255	0.298	0.324	0.365	0.364	0.354	0.374	0.366	0.334	0.321	0.275	0.228
	taud		2.202	2.283	2.334	2.335	2.407	2.457	2.415	2.436	2.514	2.460	2.285	2.112
	Ebn at Noon		167	223	256	264	274	279	269	262	253	216	158	126
	Edh at Noon		13	21	27	33	33	32	33	30	23	18	12	9
All-Sky Solar Radiation	RadAvg		89	288	760	1202	1526	1746	1559	1224	812	361	117	57
	RadStd		15	54	82	156	153	122	208	115	97	70	32	6

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page